20.3: Cloned organisms and stem cells are useful for basic research and other applications

Along with advances in DNA technology, scientists have been developing and refining methods for cloning whole multicellular organisms from single cells. In this context, cloning produces one or more organisms that are genetically identical to the "parent" that donated the single cell. This is often called *organismal cloning* to differentiate it from gene cloning and, more significantly, from cell cloning—the division of an asexually reproducing cell such as a bacterium into a group of genetically identical cells. (The common theme is that the product is genetically identical to the parent.) The current interest in organismal cloning is primarily due to its ability to generate stem cells. A **stem cell** is a relatively unspecialized cell that can both reproduce itself indefinitely and, under appropriate conditions, differentiate into specialized cells of one or more types. Stem cells have great potential for regenerating damaged tissues.

The cloning of plants and animals was first attempted over 50 years ago in experiments designed to answer basic questions about the genetic potential of single cells. For example, researchers wondered if all the cells of an organism have the same genes or whether cells lose genes during the process of differentiation (see Concept 18.4). One way to answer this question is to see whether a differentiated cell can generate a whole organism—in other words, whether cloning an organism is possible. Let's discuss these early experiments before we consider more recent progress in organismal cloning and procedures for producing stem cells.

Cloning Plants: Single-Cell Cultures

The successful cloning of whole plants from single differentiated cells was accomplished at Cornell University during the 1950s by F. C. Steward and his students, who worked with carrot plants. They found that differentiated cells taken from the root (the carrot) and incubated in culture medium could grow into normal adult plants, each genetically identical to the parent plant. These results showed that differentiation does not necessarily involve irreversible changes in the DNA. In plants, mature cells can "dedifferentiate" and then give rise to all the specialized cell types of the organism; any cell with this potential is said to be **totipotent**.

Plant cloning is used extensively in agriculture. For plants such as orchids, cloning is the only commercially practical means of producing new plants. In other cases, cloning has been used to reproduce a plant with valuable characteristics, such as resistance to plant pathogens. In fact, you yourself may be a plant cloner: If you have ever grown a new plant from a cutting, you have practiced cloning!

Cloning Animals: Nuclear Transplantation

Differentiated cells from animals generally do not divide in culture, much less develop into the multiple cell types of a new organism. Therefore, early researchers had to use a different approach to answer the question of whether differentiated animal cells are totipotent. Their approach was to remove the nucleus of an egg (creating an *enucleated* egg) and replace it with the nucleus of a differentiated cell, a procedure called *nuclear transplantation*, now more commonly called *somatic cell nuclear transfer*. If the nucleus from the differentiated donor cell retains its full genetic potential, then it should be able to direct development of the recipient cell into all the tissues and organs of an organism. Such experiments were conducted on one species of frog (*Rana pipiens*) by Robert Briggs and Thomas King in the 1950s and on another frog species (*Xenopus laevis*) by John Gurdon in the 1970s **(Figure 20.16)**. These researchers transplanted a nucleus from an embryonic or tadpole cell into an enucleated egg of the same species. In Gurdon's experiments, the transplanted nucleus was often able to support normal development of the egg into a tadpole. However, he found that the potential of a transplanted nucleus to direct normal development was inversely related to the age of the donor: The older the donor nucleus, the lower the percentage of normal tadpoles.

Figure 20.16

Inquiry: Can the nucleus from a differentiated animal cell direct development of an organism?

Experiment

John Gurdon and colleagues at Oxford University, in England, destroyed the nuclei of frog (*Xenopus laevis*) eggs by exposing the eggs to ultraviolet light. They then transplanted nuclei from cells of frog embryos and tadpoles into the enucleated eggs.

Overview of John Gurdon and colleagues' experiment.

Donor nucleus from a less differentiated cell from a frog embryo transplanted into the enucleated egg cell. Result: most develop into tadpoles. When a donor nucleus from a fully differentiated (intestinal) cell from a frog tadpole is transplanted into the enucleated egg cell most stop developing before the tadpole stage.

Results

When the transplanted nuclei came from an early embryo, the cells of which are relatively undifferentiated, most of the recipient eggs developed into tadpoles. But when the nuclei came from the fully differentiated intestinal cells of a tadpole, fewer than 2% of the eggs developed into normal tadpoles, and most of the embryos stopped developing at a much earlier stage.

Conclusion

The nucleus from a differentiated frog cell can direct development of a tadpole. However, its ability to do so decreases as the donor cell becomes more differentiated, presumably because of changes in the nucleus.

Data from J. B. Gurdon et al., The developmental capacity of nuclei transplanted from keratinized cells of adult frogs, *Journal of Embryology and Experimental Morphology*34:93–112 (1975).

WHAT IF? *If each cell in a four-cell embryo was already so specialized that it was not totipotent, what results would you predict for the experiment on the left side of the figure?*

*For suggested answer, see*Appendix A*.*

HHMI BioInteractive Video

Somatic Cell Nuclear Transfer

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

From these results, Gurdon concluded that something in the nucleus *does* change as animal cells differentiate. In frogs and most other animals, nuclear potential tends to be restricted more and more as embryonic development and cell differentiation progress. These were foundational experiments that ultimately led to stem cell technology, and Gurdon received the 2012 Nobel Prize in Medicine for this work.

Reproductive Cloning of Mammals

In addition to cloning frogs, researchers were able to clone mammals using early embryonic cells as a source of donor nuclei. Until several decades ago, though, it was not known whether a nucleus from a fully differentiated cell could be reprogrammed successfully to act as a donor nucleus. In 1997, researchers in Scotland announced the birth of Dolly, a lamb cloned from an adult sheep by nuclear transfer from a differentiated mammary gland cell **(Figure 20.17)**. Using a technique related to that in Figure 20.16, the

researchers implanted early embryos into surrogate mothers. Out of several hundred embryos, one successfully completed normal development, and Dolly was born, a genetic clone of the nucleus donor. By the age of 6, Dolly had developed premature arthritis, and complications from a lung infection led to her euthanization. This led to speculation that this sheep's cells were in some way not quite as healthy as those of a normal sheep, possibly reflecting incomplete reprogramming of the original transplanted nucleus. Reprogramming involves epigenetic changes that lead to changes in chromatin structure (see Concept 18.2), to be discussed shortly.

Figure 20.17

```
Reproductive cloning of a mammal by nuclear transfer.
```

Dolly's surrogate mother has a black face and black patches on her front legs; Dolly has a white face and no colored patches on her front legs.

Dolly, shown here as a lamb, has a very different appearance from her surrogate mother, standing beside her.

Since that time, researchers have cloned many other mammals, including mice, cats, cows, horses, pigs, and dogs. In 2018, Chinese biologists reported the first cloning of a primate, the long-tailed macaque, using the same technique that was used to clone Dolly the sheep. Then, in 2019, a different team of researchers in China announced that they had been able to culture macaque embryos in the lab, without using a surrogate mother, for 20 days after fertilization. In this case, the aim was to be able to observe early embryonic events that cannot normally be seen. However, in most cases, the research goal has been the production of new individuals, known as *reproductive cloning*.

We have learned from such experiments that cloned animals of the same species are not always identical. For example, in 2016, scientists examined four 7- to 9-year-old clones genetically identical to Dolly and concluded that, unlike Dolly, they were healthy and aging normally. Another example of nonidentity in clones is the first cloned cat, named CC for Carbon Copy **(Figure 20.18)**. The color and pattern of her calico coat differed from that of her single female parent because of random X chromosome inactivation, which is a normal occurrence during embryonic development (see Figure 15.8). And identical human twins, which are naturally occurring "clones," are always slightly different. Clearly, environmental influences and random phenomena play a significant role during development.

Figure 20.18

```
    CC (“Carbon Copy”), the first cloned cat (right), and her
single parent.
```

A cat with orange and dark patches, and a cat with white and gray patches.

Rainbow (left) donated the nucleus in a cloning procedure that resulted in CC. However, the two cats are not identical: Rainbow is a classic calico cat with orange patches on her fur and has a “reserved personality,” while CC has a gray and white coat and is more playful.

Epigenetic Differences in Cloned Animals

In most nuclear transplantation studies thus far, only a small percentage of cloned embryos develop normally to birth. And like Dolly, many cloned animals exhibit defects. Cloned mice, for instance, are prone to obesity, pneumonia, liver failure, and premature death. Scientists assert that even cloned animals that appear normal are likely to have subtle defects.

Researchers have uncovered some reasons for the low efficiency of cloning and the high incidence of abnormalities. In the nuclei of fully differentiated cells, a small subset of genes is turned on and expression of the rest of the genes is repressed. This regulation often is the result of epigenetic changes in chromatin, such as acetylation of histones or methylation of DNA (see Figure 18.7). During the nuclear transfer procedure, many of these changes must be reversed in the later-stage nucleus from a donor animal for genes to be expressed or repressed appropriately in earlier stages of development. Researchers have found that the DNA in cells from cloned embryos, like that of differentiated cells, often has abnormally high numbers of methyl groups. This finding suggests that the reprogramming of donor nuclei requires more accurate and complete chromatin restructuring than occurs during cloning procedures. Because DNA methylation helps regulate gene expression, misplaced or extra methyl groups in the DNA of donor nuclei may interfere with the pattern of gene expression necessary for normal embryonic development. In fact, the success of a cloning attempt may depend in large part on whether or not the chromatin in the donor nucleus can be artificially modified to resemble that of a newly fertilized egg.

Stem Cells of Animals

Progress in cloning mammalian embryos, including primates, has heightened speculation about the cloning of humans, which has not yet been achieved past very early embryonic stages. The main reason researchers have been trying to clone human embryos is not for reproduction, but for the production of stem cells to treat human diseases. Recall that a stem cell is a relatively unspecialized cell that can both reproduce itself

indefinitely and, under appropriate conditions, differentiate into specialized cells of one or more types **(Figure 20.19)**.

Figure 20.19

```
How stem cells maintain their own population and generate differentiated cells.
```

A stem cell can divide into another stem cell and a precursor cell. A precursor cell can differentiate into one of several cell types, depending on external factors.

Step 1: A stem cell can divide into another stem cell and a precursor cell (or into two stem cells or two precursor cells).

Step 2: A precursor cell can differentiate into one of several cell types, depending on external factors. This example is from a stem cell derived from bone marrow and it is divided into fat cells, bone cells, or white blood cells.

HHMI BioInteractive Animation

Cultured Human Embryonic Stem Cells

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

HHMI BioInteractive Animation

Somatic Cell Nuclear Transfer

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Embryonic and Adult Stem Cells

Many early animal embryos contain stem cells capable of giving rise to differentiated cells of any type. Stem cells can be isolated from early embryos at a stage called the blastula stage or its human equivalent, the blastocyst stage. In culture, these *embryonic stem (ES) cells* reproduce indefinitely, and depending on culture conditions, they can be made to differentiate into a wide variety of specialized cells **(Figure 20.20)**, including even eggs and sperm.

Figure 20.20

```
Working with stem cells.
```

Embryonic stem cells come from early human embryo (blastocyst) and can generate all embryonic cell types. Adult stem cells can generate a limited number of cell types.

Animal stem cells, which can be isolated from early embryos or adult tissues and grown in culture, are self-perpetuating, relatively undifferentiated cells. Embryonic stem cells are easier to grow than adult stem cells and can theoretically give rise to all types of cells in an organism. The range of cell types that can arise from adult stem cells is not yet fully understood.

Embryonic stem cells are from early human embryo at blastocyst stage. These cells can generate all embryonic cell types, including liver cells, nerve cells, and blood cells. Adult stem cells from bone marrow in this example are cells that can generate a limited number of cell types (blood cells shown).

HHMI BioInteractive Animation

Creating Embryonic Stem Cell Lines

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

The adult body also has stem cells, which serve to replace nonreproducing specialized cells as needed. In contrast to ES cells, *adult stem cells* are not able to give rise to all cell types in the organism, though they can generate several defined types. For example, one of the several types of stem cells in bone marrow can generate all the different kinds of blood cells (see Figure 20.20), and another type of bone marrow stem cell

can differentiate into bone, cartilage, fat, muscle, and the linings of blood vessels. To the surprise of many, the adult brain has been found to contain stem cells that continue to produce certain kinds of nerve cells there. Researchers have also reported finding stem cells in skin, hair, eyes, and dental pulp. Although adult animals have only tiny numbers of stem cells, scientists are learning to identify and isolate these cells from various tissues and, in some cases, to grow them in culture. With the right culture conditions (for instance, the addition of specific growth factors), cultured stem cells from adult animals have been made to differentiate into various defined types of specialized cells, although none are as versatile as ES cells.

Research with embryonic or adult stem cells is a source of valuable data about differentiation and has enormous potential for medical applications. The ultimate aim is to supply cells for the repair of damaged or diseased organs: for example, insulin-producing pancreatic cells for people with type 1 diabetes or certain kinds of brain cells for people with Parkinson's disease or Huntington's disease. Adult stem cells from bone marrow have long been used in bone marrow transplants as a source of immune system cells in patients whose own immune systems are nonfunctional because of genetic disorders or radiation treatments for cancer.

The developmental potential of adult stem cells is limited to certain tissues. ES cells hold more promise than adult stem cells for most medical applications because ES cells are **pluripotent**, capable of differentiating into many different cell types. One source of ES cells, first reported in 2013, is cloned human blastocysts produced by transferring a nucleus from a differentiated cell into an enucleated egg (similar to cloning Dolly the sheep). Prior to that report, cells were obtained only from embryos donated by patients undergoing infertility treatments or from long-term cell cultures originally established with cells isolated from donated embryos, an issue that prompts ethical and political discussions. Although the techniques for cloning early human embryos are still being optimized and the ethics of this procedure are being actively discussed, cloned embryos represent a potential new source for ES cells that may be less controversial. Furthermore, with a donor nucleus from a person with a particular disease, researchers should be able to produce ES cells that match the patient and are thus not rejected by his or her immune system when used for treatment. When the main aim of cloning is to produce ES cells to treat disease, the process is called *therapeutic cloning*. Although most people believe that reproductive cloning of humans is unethical, opinions vary about the morality of therapeutic cloning.

This debate has intensified with a recent result reported by same research team in China that previously cloned macaque embryos in the laboratory. In 2021, these researchers incorporated human stem cells into 6-day-old macaque embryos, creating so-called *chimeric embryos*—embryos formed by cells of two different species. Most of these chimeric embryos died, but a few (3 out of 91) persisted and had about 5% human cells. The research team described the possibility that this technology could lead to growing human organs in monkeys as a rationale for doing these experiments, but other scientists have been quite troubled by the ethical questions raised by this procedure.

Induced Pluripotent Stem (iPS) Cells

In another approach to producing stem cells for research and therapy, researchers succeeded in 2007 in learning how to turn back the clock in fully differentiated cells, reprogramming them to act like ES cells. Differentiated cells can be transformed into a type of ES cell by using a modified retrovirus to introduce extra, cloned copies of four "stem cell" master regulatory genes. The "deprogrammed" cells are known as *induced pluripotent stem* (*iPS*) cells because, in using this fairly simple laboratory technique to return them to their undifferentiated state, pluripotency has been restored. The experiments that first transformed human differentiated cells into iPS cells are described in **Figure 20.21**. Shinya Yamanaka received the 2012 Nobel Prize in Medicine for this work, shared with John Gurdon, whose work you read about in Figure 20.16.

Figure 20.21

Inquiry: Can a fully differentiated human cell be "deprogrammed" to become a stem cell?

Experiment

Shinya Yamanaka and colleagues at Kyoto University, in Japan, used a retroviral vector to introduce four genes into fully differentiated human skin fibroblast cells. The cells were then cultured in a medium that would support growth of stem cells.

> Four stem cell master regulator genes (Oct 3, 4, Sox 2, K l f 4, and c-M y c) were introduced to a skin fibroblast cell, using the retroviral cloning vector. This induced the pluripotent stem (i P S) cell.

Stem cell goes to a precursor cell and then to a skin fibroblast cell. Four stem cell master regulator genes (Oct 3, 4, Sox 2, K l f 4, and c-M y c) were introduced to the skin fibroblast cell, using the retroviral cloning vector. This induced the pluripotent stem (i P S) cell.

Results

Two weeks later, the cells resembled embryonic stem cells in appearance and were actively dividing. Their gene expression patterns, gene methylation patterns, and other characteristics were also consistent with those of embryonic stem cells. The iPS cells were able to differentiate into heart muscle cells, as well as other cell types.

Conclusion

The four genes induced differentiated skin cells to become pluripotent stem cells, with characteristics of embryonic stem cells.

Data from K. Takahashi et al., Induction of pluripotent stem cells from adult human fibroblasts by defined factors, *Cell* 131:861–872 (2007).

WHAT IF? *Patients with diseases such as heart disease or Alzheimer's could have their own skin cells reprogrammed to become iPS cells. Once affordable procedures have been developed for this and for converting iPS cells into heart or nervous system cells, the patients' own iPS cells might be used to treat their disease. When organs are transplanted from a donor to a recipient, the recipient's immune system may reject the transplant, a dangerous condition. Would using iPS cells be expected to carry the same risk? Why or why not? Given that these cells are actively dividing, undifferentiated cells, what risks might this procedure carry?*

*For suggested answer, see*Appendix A.**

By many criteria, iPS cells can perform most of the functions of ES cells, but there are some differences in gene expression and other cellular functions, such as cell division. At least until these differences are fully understood, the study of ES cells will continue to make important contributions to the development of stem cell therapies. (In fact, it is likely that ES cells will always be a focus of basic research as well.) In the meantime, work is proceeding using the iPS cells that have been experimentally produced.

There are two major potential uses for human iPS cells. First, cells from patients with particular diseases have been reprogrammed to become iPS cells, which act as model cells for studying the disease and potential treatments. Human iPS cell lines have already been developed from individuals with type 1 diabetes, Parkinson's disease, Huntington's disease, Down syndrome, and many other diseases. Second, in the field of regenerative medicine, a patient's own cells could be reprogrammed into iPS cells and then used to replace nonfunctional tissues, such as cells of the retina of the eye that have been damaged by a condition called age-related macular degeneration (AMD). Because the eye is easily accessible to treatment, while being fairly well isolated from the immune and circulatory systems, AMD and other sight disorders have been a frequent target of stem cell therapy. For instance, in the last decade, researchers have made iPS cells from skin or other cells from patients with AMD, caused them to differentiate into retinal cells, and implanted them into the patients' retinas. Thus far, there have been mixed results, depending in part on the source of the iPS cells. At best, deterioration of the patient's vision has been stopped, but some patients have experienced further loss of vision caused by factors associated with the procedure. Whether or not this approach will succeed, and be affordable enough for standard treatment, is still unknown.

Over time, the procedure may become less expensive, and some researchers have suggested creating a bank for storage of donor iPS cells that could be matched to patients' tissues. Development of techniques that direct iPS cells to become specific cell types for regenerative medicine is an area of intense research. If the challenges are met, iPS cells created in this way could eventually provide tailor-made "replacement" cells for patients without using any human eggs or embryos, thus circumventing most ethical objections.